Turnback Terminations – ¾" - 1¼"

Turnback eye terminations using 3/4" through 1" National S-506 Sleeve utilize open cavity of the S-505 Standard Steel Sleeve die on the National Hydraulic Swaging Machine. Each 3/4" through 1" S-505 Standard Steel Sleeve die contains 2 cavities (open and tapered). The 1-1/4" size S-506 sleeve utilizes the 1-3/8" socket (S-501 and S-502) die. (See page 48 for smaller sizes)

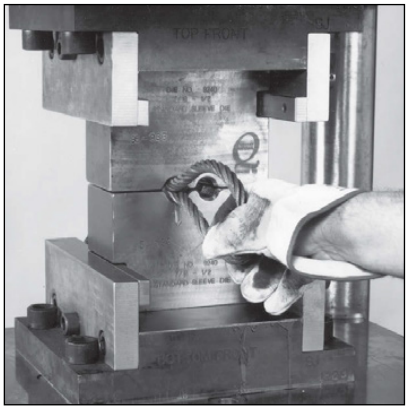
To form the eye and swage the termination, use the following procedure:

Step 1: Slide the end of the rope through the sleeve and turn back the wire rope end through the S-506 Duplex Steel Sleeve and adjust the rope to form proper eye size. Allow approximately 1 (one) rope diameter to extend past the end of the sleeve. This will provide for elongation of the fitting during swaging. **See page 47 for proper die selection.**

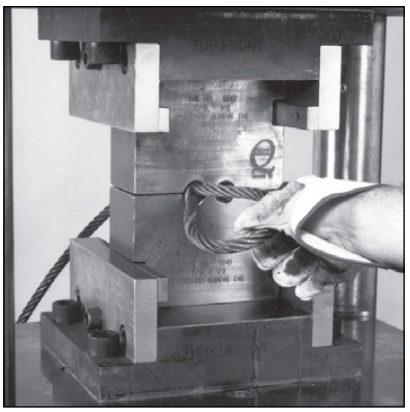


Step 2: Lubricate both die cavities and lower die face.

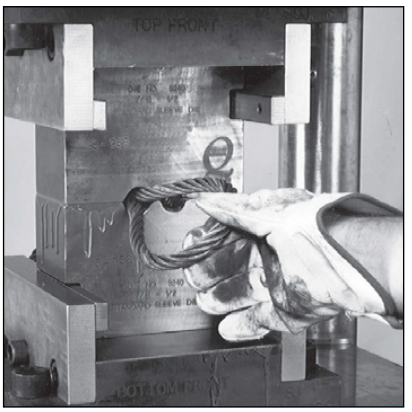
Step 3: 1st pass – allow dies to remain open approximately 1/2 the distance from the time initial contact is made between the sleeve and dies. Open dies and rotate fitting 45 to 90 degrees.



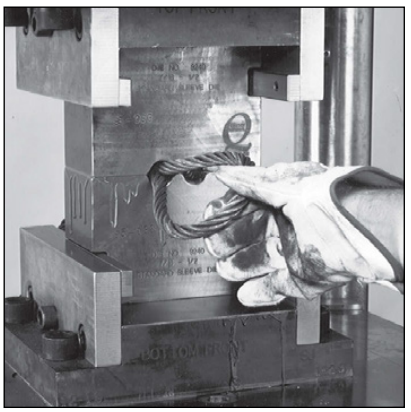
Step 4: 2nd pass – allow dies to remain open approximately 1/2 the distance from the time initial contact is made between the sleeve and dies. Open dies and rotate fitting 45 to 90 degrees.



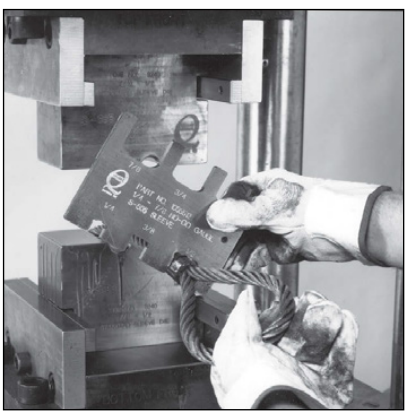
Step 5: 3rd pass – allow dies to remain open approximately 1/2 the distance from the time initial contact is made between the sleeve and dies. Open dies and rotate fitting 45 to 90 degrees.



Step 6: 4th pass – If sharp flashing does not occur, close dies. Open dies and rotate fitting 45 to 90 degrees.



Step 7: 5th and 6th pass – Close dies to round fitting. Open dies and rotate 45 to 90 degrees between passes. **Open dies and check for proper after swage dimensions.** (See Page 17 for proper use of No-Go Gauges).



Additional swaging may be required to provide smooth finish and achieve proper after swage dimensions.
Dies must be fully closed before swaging is complete.

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Swage Fittings – Hooks

Swaging Methods for S-319SWG

National S-319SWG carbon steel shank hooks are designed for swaging in National dies. These swage fittings are designed to be used on one (1) part of 6 X 19 or 6 X 37 IPS or XIP (EIP), RRL, FC, or IWRC wire rope.

Before using any National Swage fitting with any other type lay, construction or grade of wire rope, it is recommended that the termination be destructive tested and documented to prove the adequacy of the assembly to be manufactured.

Cold Swaging Process

Cold swaging is a process which causes considerable movement and work hardening of the steel in the fitting. Exerting extreme pressures, the swaging machine forces the steel to flow into the crevices between the wires and strands of the wire rope. At the same time, the swaging machine forces the material to flow plastically along the length of the wire rope. These are a few of the reasons swaging is done in multiple passes in a well lubricated die.

Protecting Dies and Swaging Machine

When the die lands touch each other, effective swaging stops. Additional swaging adds needless wear and strain on the dies and swaging machine. We recommend placing oil on the die faces and cavities.

When swaging, the oozing of the oil from the die faces will indicate when the dies are closed. At this point, stop swaging.

Avoid Excess Flashing

Flashing occurs when socket material flows out into the area between the die faces . Once formed, additional passes will not eliminate flashing. They will only fold back the flash and develop a permanent mark in the material and possibly create a crack.

To avoid excess flashing, do the following:

- Select proper die.
- Make sure that dies are in good condition and properly lubricated.
- Apply lubricant such as a light hydraulic oil to both die cavities and the lower die face.
- Follow swaging instructions on pages 54-55.



CORRECT **EXCESS FLASHING**

S-319SWG



Identification Stamping

For identification purposes, National swage products can be impression stamped after swaging. Properly done, there should be no concern for fractures initiating from such impressions.

To stamp S-319SWG hooks:

- Use round corner or low stress stamps. Stamp to maximum depth of 0.015 in. (1/64”).
- Stamp in the area on the side of the socket head, as shown in the illustration.



NOTE: DO NOT STAMP ON SWAGING SHANK.

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Shank Hooks for Swaging

"QT" QUENCHED & TEMPERED®

QUIC-CHECK®

SEE APPLICATION AND WARNING INFORMATION

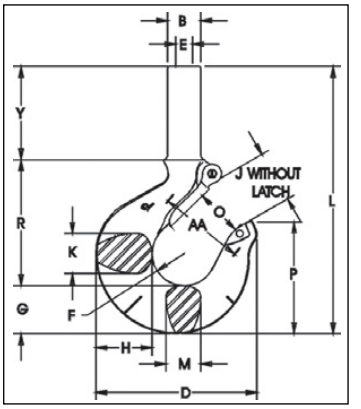
Para Español: www.thecrosbygroup.com

On Page 51 - 52

S-319SWG



- Wide range of sizes available:
  - Working Load Limit: 0.4-14 Ton
  - Wire Rope sizes: 3/16" through 1-1/8".
- Swage shank hook terminations have an efficiency rating of 95% based on the catalog strength of wire rope.
- Quenched and Tempered. Heat treat process allows for ease of swaging.
- Forged Carbon Steel.
- Design Factor of 5 to 1.
- Black Oxide finish on body (Shank is uncoated).
- Utilizes standard Crosby 319N shank hooks with interlocking hook tip. Each hook has a pre-drilled cam which can be equipped with a latch.
- Utilizes standard National Swage swaging dies.
- All hooks incorporate Crosby's patented **QUIC-CHECK®** markings (Angle Indicators and Throat Deformation Indicators). See Crosby General Catalog for detailed information.



NOTE: For use with 6 X 19 or 6 X 36, IPS or XIP (EIP), XXIP (EEIP), RRL, FC, or IWRC wire rope.

Before using any Crosby fitting with any other type lay, construction or grade of wire rope, it is recommended that the termination be destructive tested and documented to prove the adequacy of the assembly to be manufactured. Refer to swage socket or swage button instructions in the Wire Rope End Terminations User's Manual for proper swaging techniques.

S-319SWG Shank Hooks for Swaging

Wire Rope Size (in.)	Wire Rope Size (mm)	Hook ID Code	Working Load Limit (Tons)*	S-319SWG Stock No.	Weight Each (lbs.)	Required Swaging Die: Die Description	Required Swaging Die: Die Stock No.	Maximum After Swage Diameter (in.)
3/16	5	DC	0.4	1053002	.55	1/8" Button	1191621	.40
1/4	6-7	FC	0.7	1053011	.77	1/4" Socket	1192845	.46
5/16	8	GC	1.1	1053020	1.26	1/4" Button	1191621	.58
5/16	8	HC	1.1	1053039	1.83	3/8" Socket	1192863	.71
3/8	9-10	HC	1.6	1053048	1.80	3/8" Socket	1192863	.71
7/16	11	IC	2.1	1053057	3.63	1/2" Socket	1192881	.91
1/2	12-13	IC	2.8	1053066	3.58	1/2" Socket	1192881	.91
9/16	14-15	JC	3.5	1053075	7.37	5/8" Socket	1192907	1.16
5/8	16	JC	4.3	1053084	7.30	5/8" Socket	1192907	1.16
3/4	18	KC	6.2	1053093	12.73	3/4" Socket	1192925	1.42
7/8	20-22	LC	8.3	1053100	17.58	7/8" Socket	1192949	1.55
1	24-26	NC	11.0	1053119	31.46	1" Socket	1192961	1.80
1-1/8	28-30	OC **	14.0	1053128	53.73	1-1/8" Socket	1192989	2.05

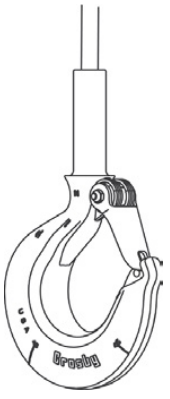
* Minimum Ultimate Load is 5 times the Working Load Limit.
** ID Code "O" is original 319 style hook.

Wire Rope Size (in.)	Wire Rope Size (mm)	S-319SWG Stock No.	Dimensions (in.) B	D	E	F	G	H	J	K	L	M	O	P	R	Y	AA
3/16	5	1053002	.44	2.86	.20	.63	.73	.81	.93	.63	5.18	.63	.93	1.96	2.39	2.00	1.50
1/4	6-7	1053011	.50	3.15	.27	.69	.84	.94	.97	.71	5.72	.71	.97	2.22	2.63	2.25	2.00
5/16	8	1053020	.65	3.59	.34	.75	1.00	1.16	1.06	.88	6.39	.88	1.06	2.44	2.80	2.50	2.00
5/16	8	1053039	.77	3.99	.34	.81	1.14	1.31	1.19	.94	7.18	.94	1.16	2.78	3.21	2.75	2.00
3/8	9-10	1053048	.77	3.99	.41	.81	1.14	1.31	1.19	.94	7.18	.94	1.16	2.78	3.21	2.75	2.00
7/16	11	1053057	.98	4.84	.48	1.00	1.44	1.63	1.50	1.31	8.70	1.13	1.41	3.47	3.92	3.25	2.50
1/2	12-13	1053066	.98	4.84	.55	1.00	1.44	1.63	1.50	1.31	8.70	1.13	1.41	3.47	3.92	3.25	2.50
9/16	14-15	1053075	1.25	6.27	.61	1.25	1.82	2.06	1.78	1.66	10.51	1.44	1.69	4.59	4.86	3.75	3.00
5/8	16	1053084	1.25	6.27	.67	1.25	1.82	2.06	1.78	1.66	10.51	1.44	1.69	4.59	4.86	3.75	3.00
3/4	18	1053093	1.55	7.54	.80	1.50	2.26	2.63	2.41	1.88	12.63	1.63	2.22	5.25	6.00	4.25	4.00
7/8	20-22	1053100	1.70	8.33	.94	1.63	2.60	2.94	2.62	2.19	13.60	1.94	2.41	5.69	6.51	4.38	4.00
1	24-26	1053119	1.98	10.38	1.06	2.13	3.01	3.50	3.41	2.69	16.80	2.38	3.19	6.88	8.30	5.38	4.00
1-1/8	28-30	1053128	2.25	13.63	1.19	2.50	3.62	4.62	4.00	3.00	23.09	3.00	3.25	8.78	9.43	9.75	6.50

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CROSBY® SHANK HOOKS FOR SWAGING

WARNINGS AND APPLICATION INSTRUCTIONS

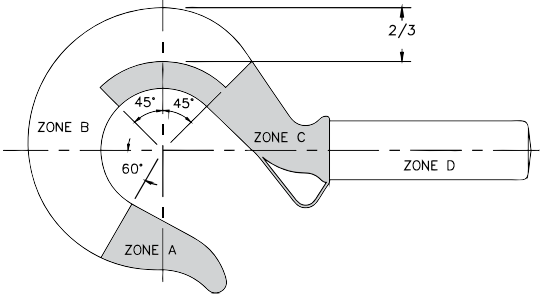


S-319SWG

- S-319SWG hooks are recommended for use with 6 x 19 or 6 x 36, IPS or XIP (EIP), XXIP (EEIP), RRL, FC or IWRC wire rope. Before using any National Swage fitting with any other type lay, construction of grade of wire rope, it is recommended that the termination be destructive tested and documented to prove the adequacy of the assembly to be manufactured.
- Use only Crosby shank hooks designed exclusively for swaging.
- A visual periodic inspection for cracks, nicks, wear gouges and deformation as part of a comprehensive documented inspection program, should be conducted by trained personnel in compliance with the schedule in ANSI B30.10.
- For hooks used in frequent load cycles or pulsating loads, the hook should be periodically inspected by Magnetic Particle or Dye Penetrant.
- Never use a hook whose throat opening has been increased, or whose tip has been bent more than 10 degrees out of plane from the hook body, or is in any other way distorted or bent.
- Note: A latch will not work properly on a hook with a bent or worn tip.
- Never use a hook that is worn beyond the limits shown in Figure 1.
- Remove from service any hook with a crack, nick, or gouge. Hooks with a nick or gouge shall be repaired by grinding lengthwise, following the contour of the hook, provided that the reduced dimension is within the limits shown in Figure 1. Contact Crosby Engineering to evaluate any crack.

⚠ WARNING

- **Loads may disengage from hook if proper procedures are not followed.**
- **A falling load may cause serious injury or death.**
- **See OSHA Rule 1926.550(g) for personnel hoisting by cranes or derricks. A Crosby 319 hook with a PL Latch attached (when secured with bolt, nut and pin) may be used for lifting personnel. A Crosby S-319N hook with an S-4320 Latch attached (when secured with cotter pin or bolt, nut and pin) may be used for lifting personnel.**
- **Hook must always support the load. The load must never be supported by the latch.**
- **Never exceed the Working Load Limit (WLL) of the wire rope and hook system.**
- **Read and understand Wire Rope End Terminations User's Manual before swaging the hook.**



ZONE A: REPAIR NOT REQUIRED
ZONE B: 10% OF ORIGINAL DIMENSION
ZONE C: 5% OF ORIGINAL DIMENSION
ZONE D: SEE MINIMUM THREAD SIZE CHART

Figure 1

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Never repair, alter, rework, or reshape a hook by welding, heating, burning, or bending.

- Never side load, back load, or tip load a hook. (See Figure 2).
- The use of a latch may be mandatory by regulations or safety codes; e.g., OSHA, MSHA, ANSI/ASME B30, insurance, etc. (Note: When using latches, see instructions in *Understanding: The Crosby Group Product Warnings* for further information.)
- Always make sure the hook supports the load (See Figure 3). The latch must never support the load (See Figure 4).
- When placing two (2) sling legs in hook, make sure the angle from the vertical to the outermost leg is not greater than 45 degrees, and the included angle between the legs does not exceed 90 degrees* (See Figure 5).

  * For angles greater than 90 degrees, or more than two (2) legs, a master link or bolt type anchor shackle should be used to attach the legs of the sling to the hook.

- See ANSI/ASME B30.10 “Hooks” for additional information.
- In accordance with ANSI B30.9, all slings terminated by swaging shall be proof tested.
- S-319SWG hooks are designed to be a component of a system, and therefore rated based on the working limit of the system of which they are attached.
- The frame code on each S-319SWG hook is to facilitate proper latch selection only, and has no reference to the working load limit of the hook.

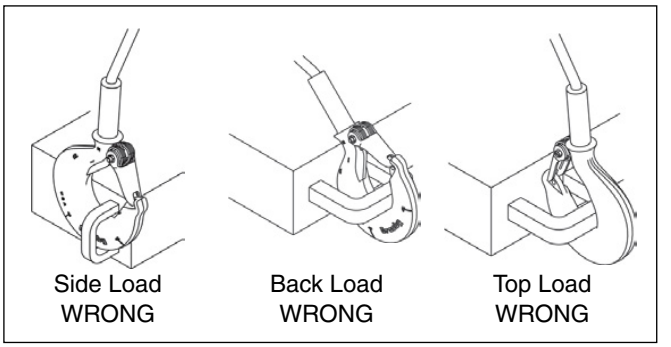


Figure 2

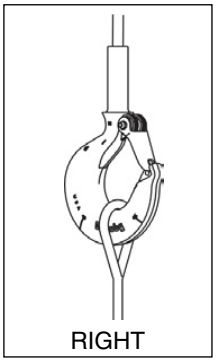


Figure 3

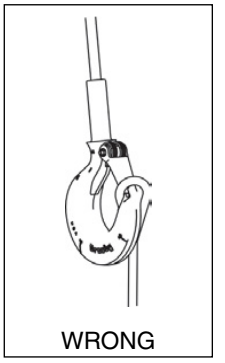


Figure 4

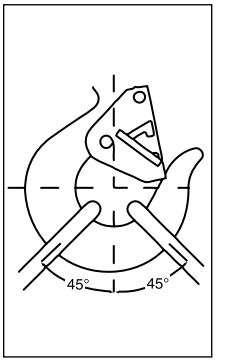


Figure 5

Wire Rope Size (in.)	Hook Frame I.D. Code	Required Swaging Die		Maximum After Swage Dimensions
		Stock No.	Description	
3/16	DC	1191621	1/8” Button	0.40
1/4	FC	1192845	1/4” Socket	0.46
5/16	GC	1191621	1/4” Button	0.46
5/16	HC	1192863	3/8” Socket	0.71
3/8	HC	1192863	3/8” Socket	0.71
7/16	IC	1192881	1/2” Socket	0.91
1/2	IC	1192881	1/2” Socket	0.91
9/16	JC	1192907	5/8” Socket	1.16
5/8	JC	1192907	5/8” Socket	1.16
3/4	KC	1192925	3/4” Socket	1.42
7/8	LC	1192949	7/8” Socket	1.55
1	NC	1192961	1” Socket	1.80
1-1/8	OC**	1192989	1-1/8” Socket	2.05

** S319C Style Hook

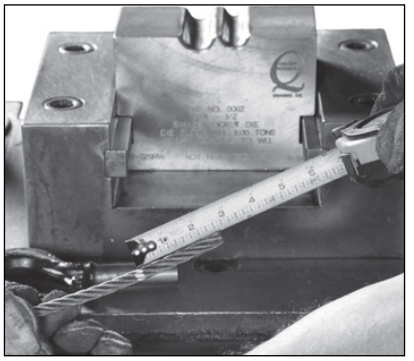
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Swage Hook Terminations – 3/16" - 1"

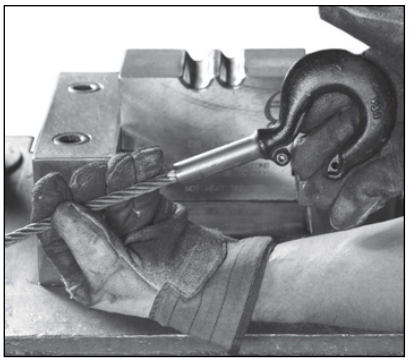
S-319SWG Swage Hooks 3/16" through 1" (See page 55 for larger sizes).

S-319SWG Swage Hooks are swaged in specially designed National Swage dies with a relief and extra length. **All hooks should be swaged full shank.** To swage the hook, use the following procedure.

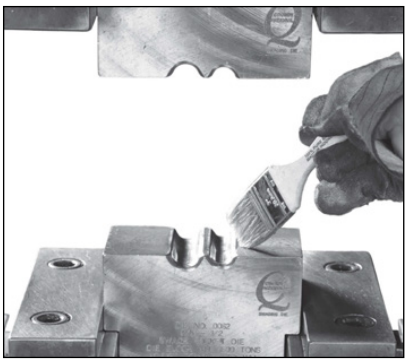
Step 1: For proper insertion, mark rope using Column Y Dimension on page 51.



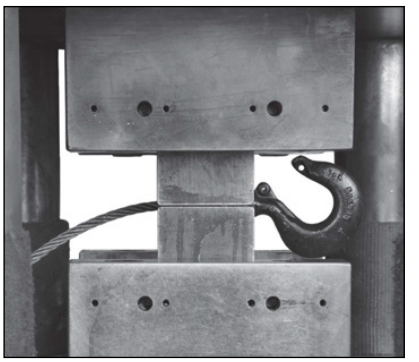
Step 2: Slide rope into hook until it is fully inserted in the shank. Inspect mark for full insertion.



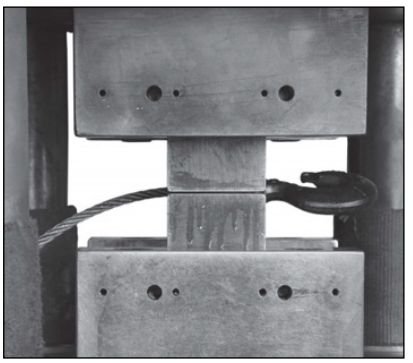
Step 3: Lubricate both die cavities and lower die face. See page 51 for proper die selection.



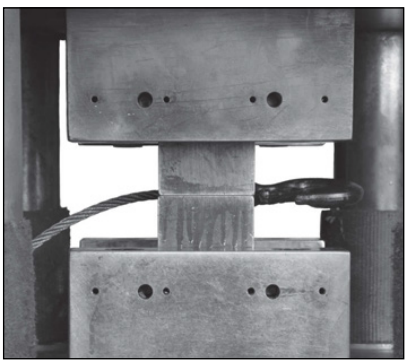
Step 4: 1st pass – allow dies to remain open approximately 1/2 the distance from time initial contact is made between hook and dies. Rotate fitting 45 to 90 degrees.

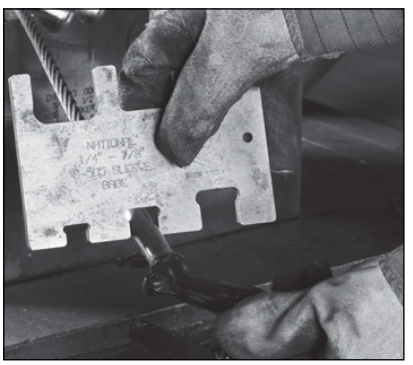


Step 5: 2nd pass – allow dies to remain open approximately ½ the distance from time initial contact is made between hook and dies. Rotate fitting 45 to 90 degrees.



Step 6: 3rd pass – If sharp flashing does not occur, close dies. Open dies and rotate fitting 45 to 90 degrees.





QUIC-CHECK®

Step 7: 4th and 5th pass – Close dies to round fitting. Open dies and rotate 45 to 90 degrees between passes. **Open dies and check termination for proper after swage dimensions.** (See page 17 for proper use of No-Go Gauges).

Additional swaging may be required to provide smooth finish and achieve proper after swage dimensions.
Dies must be fully closed before swaging is complete.

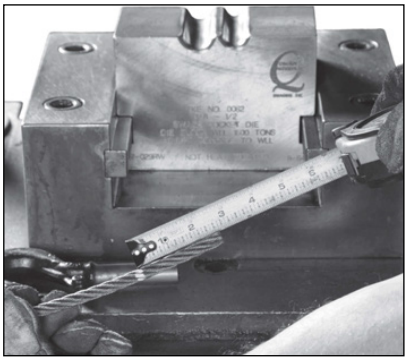
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Swage Hook Terminations – 1⅛" and Larger

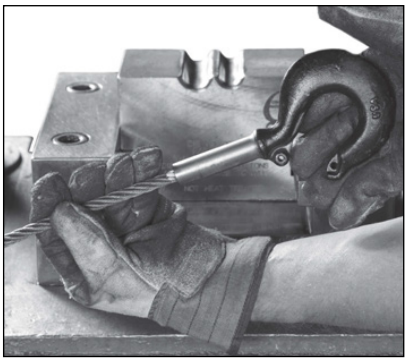
S-319SWG Swage Hooks 1-1/8" and larger (See page 54 for smaller sizes).

S-319SWG Swage Hooks are swaged in specially designed National Swage dies with a relief and extra length. **All hooks should be swaged full shank.** To swage the hook, use the following procedure.

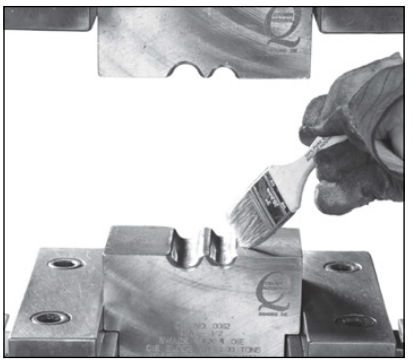
Step 1: For proper insertion, mark rope using Column Y Dimension on page 51.



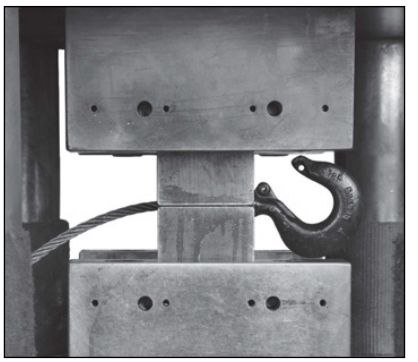
Step 2: Slide rope into hook until it is fully inserted in the shank. Inspect mark for full insertion.



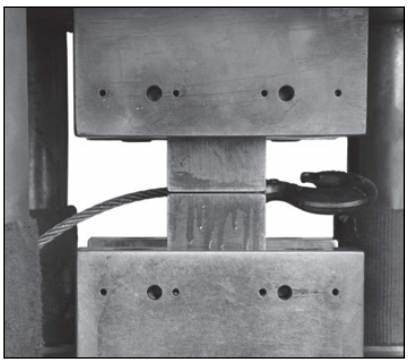
Step 3: Lubricate both die cavities and lower die face. See pages 51 for proper die selection.



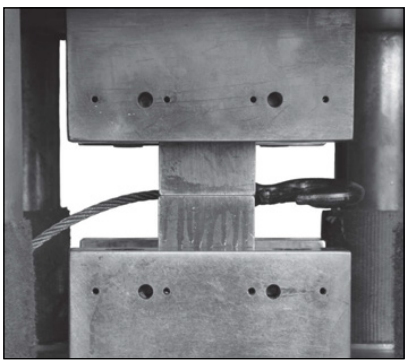
Step 4: 1st pass – allow dies to remain open approximately 1/2 the distance from time initial contact is made between hook and dies. Rotate fitting 45 to 90 degrees.

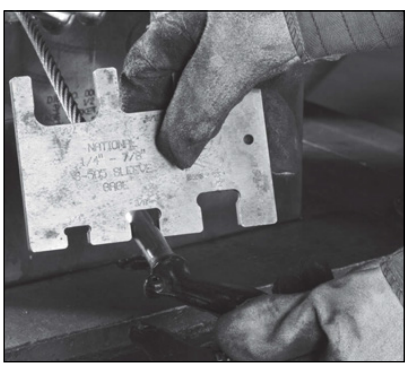


Step 5: 2nd pass – allow dies to remain open approximately ½ the distance from time initial contact is made between hook and dies. Rotate fitting 45 to 90 degrees.



Step 6: 3rd pass – If sharp flashing does not occur, close dies. Open dies and rotate fitting 45 to 90 degrees.





QUIC-CHECK®

Step 7: 4th pass – Close dies to round fitting. **Open dies and check termination for proper after swage dimensions.** (See page 17 for proper use of No-Go Gauges).

Note: Limited swaging capacity may require the use of progressive swaging (See page 41).

Additional swaging may be required to provide smooth finish and achieve proper after swage dimensions. **Dies must be fully closed before swaging is complete.**

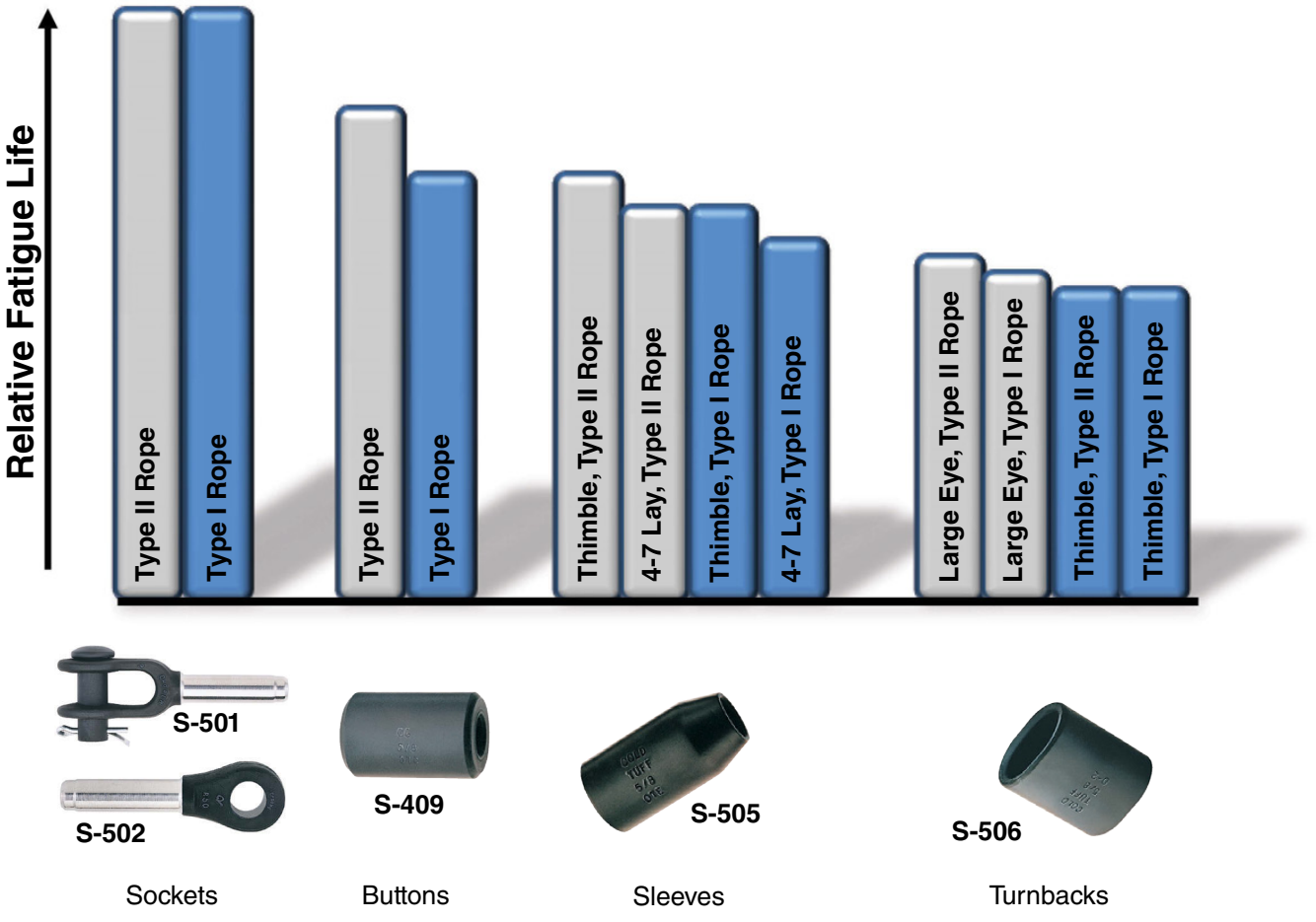
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Swage Terminal Assemblies – Fatigue Life

Relative Fatigue Life of Termination Assemblies

The chart below indicates the relative fatigue life of wire rope termination assemblies utilizing Crosby swage fittings. The findings are based upon the fatigue testing of assemblies under controlled laboratory conditions. The chart is for comparison only, not to determine the specific fatigue life values.

Refer to product section in this brochure to find the termination efficiency rating for each particular swaging product. Efficiency is calculated as a ratio of the ultimate strength of the termination assembly compared to the minimum breaking force of the wire rope. A rating is given for both Type I and Type II products.



Rope Types

Type II - 6 x 19 and 6 x 36 FC IPS RRL Wire Rope

Type I - 6 x 19 and 6 x 36 IWRC EIPS RRL Wire Rope

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Crosby® Spelter Terminations

G-416
Crosby Open
Spelter Socket



G-417
Crosby Closed
Spelter Socket



G-517
Crosby
Mooring Socket



SB-427
Crosby
Button Socket



WIRELOCK®
Socketing



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Crosby Button Spelter Sockets 65-67
WIRELOCK® 69-95

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Spelter Terminations

Application and General Information



G/S-416 Crosby Open Spelter Socket
Figure 1



G/S-417 Crosby Closed Spelter Socket
Figure 2



G-517 Crosby Mooring Socket
Figure 3



SB-427 Button Socket
Figure 4



W416-7 WIRELOCK® Socket Compound
Figure 5

There are four basic types of sockets available from Crosby that utilize a poured zinc or resin to create the termination. Poured sockets are used in many wire rope and structural strand termination applications where a high strength, permanent end attachment is required.

Crosby Spelter Sockets are available in both the closed and open design (see Figure 1 and Figure 2 above). These types of poured sockets are generally used as a straight tension member. They are commonly used on crane boom pendant lines and on applications that require an anchoring system to be connected mechanically to a structure, such as on bridges and roof systems. They are also occasionally used on crane hoist lines to connect to a crane block or overhaul ball.

Crosby Mooring Sockets (See Figure 3) are designed to be used in marine applications where attachment to shackles and connecting links is required.

Crosby Button Sockets (see Figure 4) are designed to terminate all types of running ropes used on mobile cranes to connect a hoist line to a crane block or overhaul ball.

The common thread among all these products is the method used to terminate the rope. **WIRELOCK®** (see Figure 5) is a resin compound offered by Crosby, that when properly used, creates a 100% efficient termination. Zinc is also used to terminate spelter type sockets with the same resultant efficiency.

Efficiency

Efficiency ratings for wire rope end terminations are based upon the minimum breaking force of wire rope.

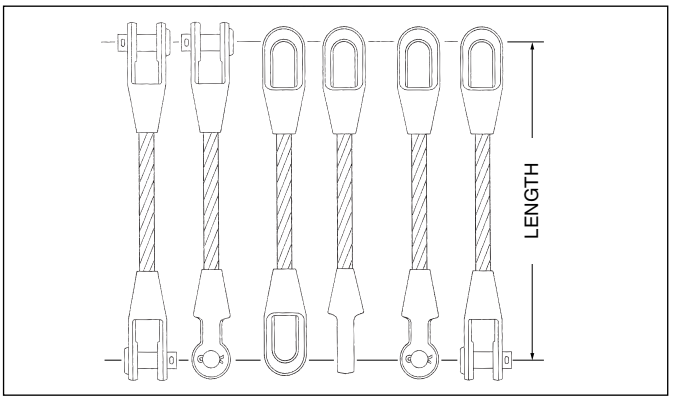
When properly fabricated, all Crosby Spelter Sockets have a termination efficiency of 100 percent based on the minimum breaking force of the wire rope (see table at top of page). Ratings for Crosby Spelter Sockets are based on recommended use with 6 x 7, 6 x 19, or 6 x 36, IPS or XIP (EIP), XXIP (EEIP), RRL, FC, or IWRC wire rope.

Strand constructed with minimal number of wires (e.g. 1 x 7) requires special consideration that socket basket length be five (5) times the strand diameter or fifty (50) times the wire diameter, whichever is greater.

Product	Figure # from Left	Efficiency
G/S-416 Open Spelter Sockets	1	100%
G/S-417 Closed Spelter Sockets	2	100%
G-517 Mooring Sockets	3	100%
SB-427 Button Sockets	4	100%

INSTALLATION INFORMATION

1. The spelter process begins by inserting the end of the wire rope into the narrow end of the cone shaped socket or button.
2. Before pouring the socket, the wire must be broomed and properly cleaned. Once the socket (or button) and wire rope are properly prepared, the broomed wire rope is pulled inside the socket cavity.
3. The socket (or button) cavity is then poured with molten zinc, or more commonly, an epoxy resin called **WIRELOCK®**.
4. The socket must then be allowed to cure based on zinc/resin manufacturer's recommendations.
5. Assembly length is measured from center line of pins for both open and closed sockets (bearing point to bearing point).



Key Facts About Spelter Terminations:

1. Poured spelter sockets are occasionally used in sling applications. However, if used in a choker hitch sling application, the rated capacity must be reduced based on angle of choke. See wire rope capacity chart for complete information.
2. Assemblies with poured sockets can have tight tolerances. Tolerances as small as plus or minus 1/8 inch is not uncommon.
3. Specifications such as fitting type, orientation of pin, whether zinc or epoxy resin is to be used and type of application should be communicated to the fabricator of the socket assembly.
4. Proper training is required before attempting to pour sockets. See proper socketing procedures in following pages for complete information.

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Spelter Terminations

Reuse of Crosby Spelter Sockets and Buttons

Crosby allows the reuse of sockets and buttons so long as recommended procedures are followed.

The following are general guidelines for the reuse of Crosby 416/417 Spelter Sockets and Crosby 427 Spelter Buttons. The use and inspection of used spelter sockets and buttons is the responsibility of the user.

Procedure for Removing Spelter Cone

1. Cut the rope close (1/2") to the nose end of the socket/button and press the cone out of the basket of the socket.
2. We do not recommend the use of heat to remove the spelter cone for metallurgical, medical and environmental reasons.
   A. However, if this is the only means available for removing the zinc cone, care should be taken not to exceed 850°F (450°C) surface temperature. The preferred method would be a slow heat in a temperature controlled oven. If a torch (rosebud) is used, the heat spot shall be monitored with a tempil stick or a temperature indicator to prevent localized heating from exceeding the 850°F (450°C) limit.
   B. To remove a **WIRELOCK®** cone, heat the surface of the socket/button to 350°F (177°C). Do not exceed the 850°F (450°C) limit for any localized hot spot. Leave for 5-10 minutes, then drive the cone out with a hammer and drift.

Selection of Sockets and Buttons for Reuse

1. Use only sockets/buttons that do not show discoloration from excessive heating or any signs of welding.
2. Select only sockets/buttons that have been cleaned and have passed a Magnetic Particle Inspection by a qualified technician (Level II ASNT-SNT-TC-1A-88) per ASTM E709. Acceptance criteria shall be per ASTM E125, Types II-VIII, Degree 1. No cracks are acceptable.
3. Select only sockets/buttons that do not show any signs of overloading or wear on the socket or pin, (ie. elongated pin holes, undersized pins, etc.).
4. Select sockets/buttons that are free from nicks, gouges and abrasions. Indications my be repaired by lightly grinding until surfaces are smooth, provided they do not reduce the dimensions by more than 10% of the nominal catalog dimension.
5. Select sockets/buttons that are not distorted, bent or deformed.

   Note: Sockets/Buttons having any of the indications as outlined above (1-5) shall not be reused.

Procedures For Speltering Sockets & Buttons

1. The proper procedure for speltering sockets can be found in the *Wire Rope User's Manual, 3rd Edition, Appendix C, American Petroleum Institute (API) Recommended Practice 9B (RP9B), 11th Edition, September, 2002* or ISO17558:2006 Steel Wire Ropes Socketing Procedures – Molten Metal and Resin Socketing.
2. Some standards (API, ISO, BSI) recommend preheating of the zinc spelter socket/button before pouring. This temperature shall not exceed 850°F (450°C).
3. Resin spelter sockets/buttons shall follow the procedure outlined by the resin manufacturer (See **WIRELOCK®**, Pages 73-77).

Proof Testing

We recommend the socketed assembly be proof tested at two (2) times the Working Load Limit (2 x WLL) assigned to the socketed assembly.

STANDARD CONSIDERATIONS

ASME B30.9

All slings terminated with sockets shall be proof loaded. The proof load shall be a minimum of 2 and a maximum of 2.5 times the single leg vertical hitch rated load. The proof load should be that specified by the wire rope or fitting manufacturer's recommendation provided that it is within the above specified proof load range. When sockets are used in sling assemblies the minimum recommended design factor is 5.

ASME B30.5

Cranes require that boom pendants terminated with sockets be proof tested. When sockets are used in boom pendant lines, the minimum recommended design factor is 3.

See OSHA and ASME for full information.

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Crosby® Spelter Sockets

Speltering Methods for G&S 416 and 417 Sockets

Crosby Spelter Sockets are available in both the closed and open design. These types of poured sockets are generally used as a straight tension member. They are commonly used on crane boom pendant lines and applications that require an anchoring system to be connected mechanically to a structure, such as on bridges and roof systems. They are also occasionally used on crane hoist lines to connect to a crane block or overhaul ball.

Efficiency

When properly fabricated, all Crosby Spelter Sockets have a termination efficiency of 100 percent based on the minimum breaking force of the wire rope (see table on page 58).

Ratings for Crosby Spelter Sockets are based on recommended use with 6 x 7, 6 x 19, or 6 x 36, IPS or XIP (EIP), XXIP (EEIP), RRL, FC, or IWRC wire rope.

Strand constructed with minimal number of wires (e.g. 1 x 7) requires special consideration that socket basket be five (5) times the strand diameter or fifty (50) times the wire diameter, whichever is the greater.

Before using any Crosby fitting with any other type lay, construction or grade of wire rope, it is recommended that the termination be destructive tested and documented to prove the adequacy of the assembly to be manufactured.

Installation Information

General Overview

1. The spelter socket is attached by inserting the end of the wire rope into the narrow end of the cone shaped socket.
2. Before pouring the socket, the wire must be broomed and properly cleaned. Once the socket and wire rope are properly prepared, the broomed wire rope is pulled into the socket cavity.
3. The socket cavity is then poured with molten zinc, or more commonly, an epoxy resin called **WIRELOCK®**.
4. The socket must then be allowed to cure based on zinc/resin manufacturer's recommendations.
5. Assembly length is measured from center line of pins for both open and closed sockets.

Specific Instructions

Socketing Using WIRELOCK® Resin Material:

Seizing, cleaning, brooming and preparation of wire rope and pouring of **WIRELOCK®** is to be carried out per instructions provided in the Wire Rope End Terminations User's Manual (see pages 73-77 in this document), and **WIRELOCK®** Warnings and Application Instructions located on the **WIRELOCK®** Product or in the Crosby® General Catalog.



Socketing using Zinc Spelter Material:

Seizing, cleaning, brooming and preparation of the wire rope, and pouring of zinc is to be carried out in accordance with recommendations of the Wire Rope User's Manual, 3rd Edition, Appendix C, American Petroleum Institute (API) Recommended Practice 9B (RP9B), 11th Edition, September, 2002 or ISO17558:2006 Steel Wire Ropes Socketing Procedures – Molten Metal and Resin Socketing.

Before operation of the wire rope assembly, it is recommended that all poured sockets, whether with zinc or resin, be proof loaded to seat the cone.

Reuse of Crosby Spelter Sockets and Buttons

Crosby allows the reuse of sockets so long as recommended procedures are followed.

Refer to Page 59 for recommended procedures for reuse of Crosby Spelter Sockets.

STANDARD CONSIDERATIONS

ASME B30.9

All slings terminated with sockets shall be proof loaded. The proof load shall be a minimum of 2 and a maximum of 2-1/2 times the single leg vertical hitch rated load. The proof load should be that specified by the wire rope or fitting manufacturer's recommendation provided that it is within the above specified proof load range. When sockets are used in sling assemblies the minimum recommended design factor is 5.

ASME B30.5

Cranes require that boom pendants terminated with sockets be proof tested. When sockets are used in boom pendant lines, the minimum recommended design factor is 3.

See OSHA and ASME for full information.

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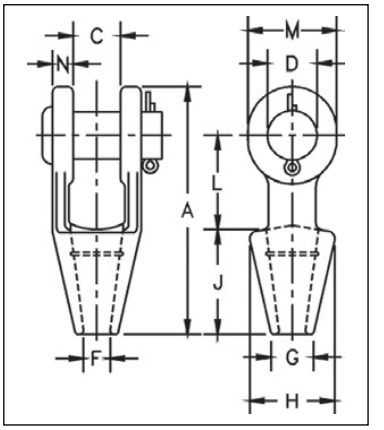
Open Spelter Sockets

G-416 / S-416



Open grooved sockets meet the performance requirements of Federal Specification RR-S-550D, Type A, except for those provisions required of the contractor. For additional information, see Crosby General Catalog.

- Forged steel sockets through 1-1/2″, cast alloy steel 1-5/8″ through 4″.
- Spelter socket terminations have an efficiency rating of 100%, based on the catalog strength of wire rope. Ratings are based on recommended use with 6 x 7, 6 x 19, or 6 x 36, IPS or XIP (EIP), XXIP (EEIP), RRL, FC, or IWRC wire rope. Strand constructed with minimal number of wires (e.g. 1 x 7) requires special consideration that socket basket be five (5) times the strand diameter or fifty (50) times the wire diameter, whichever is the greater.



NOTICE: All cast steel sockets 1-5/8″ and larger are magnetic particle inspected and ultrasonic inspected. Proof testing available on special order.

Drawing illustrates one groove used on sockets 1/4″ through 3/4″. Sizes 7/8″ through 1-1/2″ use 2 grooves. Sizes 1-5/8″ and larger use 3 grooves.

G-416 / S-416 Open Spelter Sockets

Rope Dia.		Structural Strand Dia. (in.)	Ultimate Load (t)	Stock No.		Weight Each (lbs.)	Dimensions (in.)									
(in.)	(mm)			G-416 Galv.	S-416 S.C.		A	C	D	F	G	H	J	L	M	N
1/4	6-7	—	4.5	1039619	1039628	1.10	4.56	.75	.69	.38	.69	1.56	2.25	1.56	1.31	.36
5/16-3/8	8-10	—	12	1039637	1039646	1.30	4.84	.81	.81	.50	.81	1.69	2.25	1.75	1.50	.44
7/16-1/2	11-13	—	20	1039655	1039664	2.25	5.56	1.00	1.00	.56	.94	1.88	2.50	2.00	1.88	.50
9/16-5/8	14-16	1/2	27	1039673	1039682	3.60	6.75	1.25	1.19	.69	1.13	2.25	3.00	2.50	2.25	.56
3/4	18	9/16-5/8	43	1039691	1039708	5.83	7.94	1.50	1.38	.81	1.25	2.62	3.50	3.00	2.62	.62
7/8	20-22	11/16-3/4	55	1039717	1039726	9.65	9.25	1.75	1.63	.94	1.50	3.25	4.00	3.50	3.13	.80
1	24-26	13/16-7/8	78	1039735	1039744	15.50	10.56	2.00	2.00	1.13	1.75	3.75	4.50	4.00	3.75	.88
1-1/8	28-30	15/16-1	92	1039753	1039762	21.50	11.81	2.25	2.25	1.25	2.00	4.12	5.00	4.62	4.12	1.00
1-1/4 - 1-3/8	32-35	1-1/16 - 1-1/8	136	1039771	1039780	31.00	13.19	2.50	2.50	1.50	2.25	4.75	5.50	5.00	4.75	1.13
1-1/2	38	1-3/16 - 1-1/4	170	1039799	1039806	47.25	15.12	3.00	2.75	1.63	2.75	5.25	6.00	6.00	5.38	1.19
* 1-5/8	* 40-42	1-5/16 - 1-3/8	188	1039815	1039824	55.00	16.25	3.00	3.00	1.75	3.00	5.50	6.50	6.50	5.75	1.31
* 1-3/4 - 1-7/8	* 44-48	1-7/16 - 1-5/8	268	1039833	1039842	82.00	18.25	3.50	3.50	2.00	3.13	6.38	7.50	7.00	6.50	1.56
* 2 - 2-1/8	* 50-54	1-11/16 - 1-3/4	291	1039851	1039860	129.00	21.50	4.00	3.75	2.25	3.75	7.38	8.50	9.00	7.00	1.81
* 2-1/4 - 2-3/8	* 56-60	1-13/16 - 1-7/8	360	1039879	1039888	167.00	23.50	4.50	4.25	2.50	4.00	8.25	9.00	10.00	7.75	2.13
* 2-1/2 - 2-5/8	* 64-67	1-15/16 - 2-1/8	424	1041633	1041642	252.00	25.50	5.00	4.75	2.88	4.50	9.25	9.75	10.75	8.50	2.38
* 2-3/4 - 2-7/8	* 70-73	2-3/16 - 2-7/16	511	1041651	1041660	315.00	27.25	5.25	5.00	3.12	4.88	10.50	11.00	11.00	9.00	2.88
* 3 - 3-1/8	* 75-80	2-1/2 - 2-5/8	563	1041679	1041688	380.00	29.00	5.75	5.25	3.38	5.25	11.12	12.00	11.25	9.50	3.00
* 3-1/4 - 3-3/8	* 82-86	2-3/4 - 2-7/8	722	1041697	1041704	434.00	30.88	6.25	5.50	3.62	5.75	11.88	13.00	11.75	10.00	3.12
* 3-1/2 - 3-5/8	* 88-92	3 - 3-1/8	779	1041713	1041722	563.00	33.25	6.75	6.00	3.88	6.50	12.38	14.00	12.50	10.75	3.25
* 3-3/4 - 4	* 94-102	—	875	1041731	1041740	783.00	36.25	7.50	7.00	4.25	7.25	13.62	15.00	13.50	12.50	3.50

* Cast Alloy Steel.

NOTE: Available with bolt, nut and cotter. Contact Crosby for more information.

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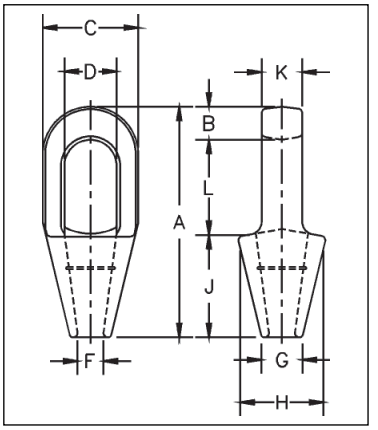
Closed Spelter Sockets

G-417 / S-417



Closed grooved sockets meet the performance requirements of Federal Specification RR-S-550D, Type B, except for those provisions required of the contractor. For additional information, see Crosby General Catalog.

- Forged steel sockets through 1-1/2", cast alloy steel 1-5/8" through 4".
- Spelter socket terminations have an efficiency rating of 100%, based on the catalog strength of wire rope. Ratings are based on the recommended use with 6 x 7, 6 x 19 or 6 x 36, IPS or XIP (EIP) , XXIP (EEIP), RRL, FC or IWRC wire rope. Strand constructed with minimal number of wires (e.g. 1 x 7) requires special consideration that socket basket be five (5) times the strand diameter or fifty (50) times the wire diameter, whichever is the greater.



NOTICE: All cast steel sockets 1-5/8" and larger are magnetic particle inspected and ultrasonic inspected. Proof testing available on special order.

Drawing illustrates one groove used on sockets 1/4" through 3/4". Sizes 7/8" through 1-1/2" use 2 grooves. Sizes 1-5/8" and larger use 3 grooves.

G-417 / S-417 Closed Spelter Sockets

Rope Dia.		Structural Strand Dia.	Ultimate Load	Stock No.		Weight Each	Dimensions (in.)									
(in.)	(mm)	(in.)	(t)	G-417 Galv.	S-417 S.C.	(lbs.)	A	B	C	D*	F	G	H	J	K	L
1/4	6-7	—	4.50	1039897	1039904	.50	4.50	.50	1.50	.88	.38	.69	1.56	2.25	.50	1.75
5/16 - 3/8	8-10	—	12.0	1039913	1039922	.75	4.94	.62	1.69	.97	.50	.81	1.69	2.25	.69	2.06
7/16 - 1/2	11-13	—	20.0	1039931	1039940	1.50	5.50	.69	2.00	1.16	.56	.94	2.00	2.50	.88	2.31
9/16 - 5/8	14-16	1/2	30.8	1039959	1039968	2.50	6.31	.81	2.63	1.41	.69	1.12	2.38	3.00	1.00	2.50
3/4	18	9/16 - 5/8	43.5	1039977	1039986	4.25	7.62	1.06	3.00	1.66	.88	1.25	2.75	3.50	1.25	3.06
7/8	20-22	11/16 - 3/4	65.3	1039995	1040000	7.25	8.75	1.25	3.63	1.94	1.00	1.50	3.25	4.00	1.50	3.50
1	24-26	13/16 - 7/8	81.6	1040019	1040028	10.50	9.88	1.38	4.13	2.30	1.13	1.75	3.75	4.50	1.75	4.00
1-1/8	28-30	15/16 -1	100	1040037	1040046	14.25	11.00	1.50	4.50	2.56	1.25	2.00	4.13	5.00	2.00	4.50
1-1/4 -1-3/8	32-35	1-1/16 -1-1/8	136	1040055	1040064	19.75	12.12	1.63	5.00	2.81	1.50	2.25	4.75	5.50	2.25	5.00
1-1/2	38	1-3/16 - 1-1/4	170	1040073	1040082	29.20	13.94	1.94	5.38	3.19	1.63	2.75	5.25	6.00	2.50	6.00
† 1-5/8	† 40-42	1-5/16 - 1-3/8	188	1040091	1040108	36.00	15.13	2.13	5.75	3.25	1.75	3.00	5.50	6.50	2.75	6.50
† 1-3/4 - 1-7/8	† 44-48	1-7/16 - 1-5/8	268	1040117	1040126	57.25	17.25	2.19	6.75	3.75	2.00	3.13	6.38	7.50	3.00	7.56
† 2 - 2-1/8	† 50-54	1-11/16 - 1-3/4	309	1040135	1040144	79.00	19.87	2.44	7.63	4.38	2.25	3.75	7.38	8.50	3.25	8.81
† 2-1/4 - 2-3/8	† 56-60	1-13/16 - 1-7/8	360	1040153	1040162	105.00	21.50	2.75	8.50	5.00	2.63	4.13	8.25	9.00	3.63	9.75
† 2-1/2 - 2-5/8	† 64-67	1-15/16 - 2-1/8	424	1041759	1041768	140.00	23.50	3.12	9.50	5.50	2.88	4.50	9.25	9.75	4.00	10.62
† 2-3/4 - 2-7/8	† 70-73	2-3/16 - 2-7/16	549	1041777	1041786	220.00	25.38	3.12	10.75	6.25	3.12	4.88	10.19	11.00	4.88	11.25
† 3 - 3-1/8	† 75-80	2-1/2 - 2-5/8	656	1041795	1041802	276.00	27.12	3.37	11.50	6.75	3.38	5.25	11.50	12.00	5.25	11.75
† 3-1/4 - 3-3/8	† 82-86	2-3/4 - 2-7/8	750	1041811	1041820	313.00	29.25	4.00	12.25	7.25	3.62	5.75	12.25	13.00	5.75	12.25
† 3-1/2 - 3-5/8	† 88-92	3 - 3-1/8	820	1041839	1041848	400.00	31.00	4.00	13.00	7.75	3.88	6.31	13.00	14.00	6.25	13.00
† 3-3/4 - 4	† 94 - 102	—	1005	1041857	1041866	542.00	33.25	4.25	14.25	8.50	4.25	7.25	14.25	15.00	7.00	14.00

* Diameter of pin must not exceed pin used on companion 416 Socket. Reference adjacent page "D" dimension.
† Cast Alloy Steel.

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Crosby® Mooring Sockets

Speltering Methods for G-517 Mooring Sockets

Crosby Mooring Sockets are designed to be used in marine applications where attachment to shackles and connecting links is required.

Efficiency

When properly fabricated, all Crosby Mooring Sockets have a termination efficiency of 100 percent based on the minimum breaking force of the wire rope (see table on page 58).

Ratings for Crosby Mooring Sockets are based on recommended use with 6 x 7, 6 x 19, or 6 x 36, IPS or XIP (EIP), XXIP (EEIP), RRL, FC, or IWRC wire rope.

Strand constructed with minimal number of wires (e.g. 1 x 7) requires special consideration that socket basket be five (5) times the strand diameter or fifty (50) times the wire diameter, whichever is the greater.

Before using any Crosby fitting with any other type lay, construction or grade of wire rope, it is recommended that the termination be destructive tested and documented to prove the adequacy of the assembly to be manufactured.

Installation Information

General Overview

1. The socket is attached by inserting the end of the wire rope into the narrow end of the cone shaped socket.
2. Before pouring the socket, the wire must be broomed and properly cleaned. Once the socket and wire rope are properly prepared, the broomed wire rope is pulled into the socket cavity.
3. The socket cavity is then poured with molten zinc, or more commonly, **WIRELOCK**® resin.
4. The socket must then be allowed to cure based on zinc/resin manufacturer's recommendations.
5. Assembly length is measured from center line of pins for both open and closed sockets.

Specific Instructions

Socketing Using WIRELOCK® Resin Material:

Seizing, cleaning, brooming and preparation of wire rope and pouring of **WIRELOCK**® is to be carried out per instructions provided in the pages 73-77 in this document, and **WIRELOCK**® Warnings and Application Instructions located on the **WIRELOCK**® Product or in the Crosby® General Catalog.



Socketing Using Zinc Spelter Material:

Seizing, cleaning, brooming and preparation of the wire rope, and pouring of zinc is to be carried out in accordance with recommendations of the Wire Rope User's Manual, 3rd Edition, Appendix C, American Petroleum Institute (API) Recommended Practice 9B (RP9B), 11th Edition, September, 2002 or ISO17558:2006 Steel Wire Ropes Socketing Procedures – Molten Metal and Resin Socketing.

Before operation of the wire rope assembly, it is recommended that all poured sockets, whether with zinc or resin, be proof loaded to seat the cone.

Reuse of Crosby Mooring Sockets

Crosby allows the reuse of sockets so long as recommended procedures are followed.

Refer to Page 59 for recommended procedures for reuse of Crosby Spelter Sockets.

Key Facts About Wire Rope Spelter Sockets:

1. Design of bail allows for easy connection to shackles and other connecting links.
2. Socket design utilizes features to keep cone from rotating.

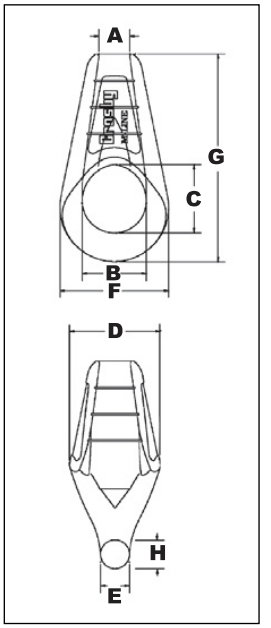
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Mooring Sockets

G-517



- Wide range of sizes available:
  - 1-1/4" thru 4" wireline
- "M-Line" socket terminations have a 100% efficiency rating, based on the catalog strength of the wire rope. Ratings are based on recommended use with 6 x 7, 6 x 19, or 6 x 36, IPS or XIP (EIP), XXIP (EEIP), RRL, FC, or IWRC wire rope. Strand constructed with minimal number of wires (e.g. 1 x 7) requires special consideration that socket basket be five (5) times the strand diameter or fifty (50) times the wire diameter, whichever is the greater.
- Galvanized finish.
- Designed for today's higher strength classes of wire rope.
- Design of bail allows for easy connection to shackles and other connecting links.
- Socket design utilizes features to keep cone from rotating.



NOTICE: All Cast Mooring Sockets are Individually Magnetic Particle Inspected and Ultrasonic Inspected.

G-517 "M-Line" Mooring Sockets

Wire Rope Size		Ultimate Load	G-517	Weight Each	Dimensions (in.)							
(in.)	(mm)	(t)	Stock No.	(lbs.)	A	B	C	D	E	F	G	H
1-1/4 - 1-3/8	32-35	113	1004943	17	1.63	3.09	3.63	4.45	1.44	5.13	10.89	1.53
1-1/2 - 1-5/8	38-41	136	1004961	30	1.95	3.69	4.32	5.43	1.60	6.31	13.00	1.81
1-3/4 - 1-7/8	44-48	181	1004989	43	2.23	4.16	4.53	6.30	1.84	7.22	14.11	2.09
2 - 2-1/8	50-54	227	1005002	57	2.50	4.75	5.26	7.02	2.09	8.25	16.02	2.24
2-1/4 - 2-3/8	57-60	277	1005020	76	2.78	5.25	5.78	7.72	2.31	9.16	17.90	2.62
2-1/2 - 2-5/8	64-67	363	1005048	106	3.05	5.88	6.71	8.53	2.69	10.13	19.89	2.66
2-3/4 - 2-7/8	70-73	454	1005066	138	3.33	6.50	7.13	9.35	3.00	11.09	21.63	2.98
3 - 3-1/8	76-79	544	1005084	193	3.50	7.25	7.74	10.30	3.25	12.31	23.50	3.24
3-1/4 - 3-3/8	82-86	635	1005105	229	3.81	7.62	8.80	10.94	3.50	13.13	25.75	3.43
3-1/2 - 3-5/8	88-92	735	1005123	279	4.15	8.00	9.06	11.72	3.69	13.96	27.70	4.12
3-3/4 - 4	95-102	907	1005141	384	4.39	8.75	10.50	12.91	3.69	15.88	30.13	4.46

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Crosby® Button Sockets

Speltering Methods for SB-427 Button Sockets

Crosby Button Sockets are designed to terminate all types of running ropes used on mobile cranes, to connect hoist line.

Efficiency

When properly fabricated, all Crosby Button Sockets have a termination efficiency of 100 percent based on the minimum breaking force of the wire rope (see table on page 58).

Ratings for Crosby Button Sockets are based on recommended use with 6 x 7, 6 x 19, or 6 x 36, IPS or XIP (EIP), XXIP (EEIP), RRL, FC, or IWRC wire rope.

Before using any Crosby fitting with any other type lay, construction or grade of wire rope, it is recommended that the termination be destructive tested and documented to prove the adequacy of the assembly to be manufactured.

Installation Information

Install button on the rope so that the live end of the rope extends out of small diameter of the button. Broomed end of rope should be inserted to the "MAX FILL" line marked on the button to ensure correct length of engagement with socketing material and allow installation of pulling eye.

Socketing Using WIRELOCK® Resin Material:

Seizing, cleaning, brooming and preparation of wire rope and pouring of **WIRELOCK®** is to be carried out per instructions provided in the pages 73-77 in this document, and **WIRELOCK®** Warnings and Application Instructions located on the **WIRELOCK®** Product or in the Crosby® General Catalog.

Socketing using Zinc Spelter Material:

Seizing, cleaning, brooming and preparation of the wire rope, and pouring of zinc is to be carried out in accordance with recommendations of Appendix C, or other approved procedures.



Before operation of the wire rope assembly, it is recommended that all poured sockets, whether with zinc or resin, be proof loaded to seat the cone.

Reuse of Crosby Button Sockets

Crosby allows the reuse of sockets so long as recommended procedures are followed.

Refer to Page 59 for recommended procedures for reuse of Crosby Spelter Sockets.

Key Facts About Wire Rope Button Sockets:

1. Can be used to terminate high performance, rotation resistant ropes and standard 6 strand ropes.
2. Easy to install assembly utilizes Crosby® **WIRELOCK®** socketing compound.
3. Sockets and buttons are reusable.
4. Replacement buttons and sockets are available.
5. Locking feature available to prevent rotation of rope.
6. Button contains cap with eye that can be attached to, and used to pull rope during reeving process.

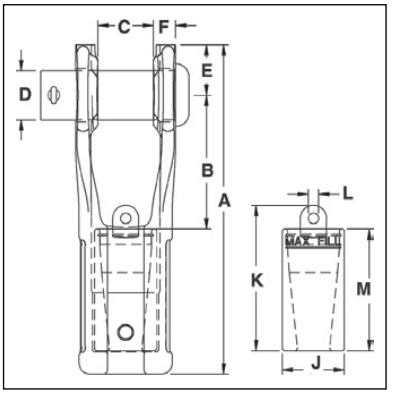
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Button Spelter Sockets

SB-427



- Available in six (6) sizes from 1/2″ to 1-1/2″, (13mm - 38mm).
- Button Spelter terminations have a 100% efficiency rating, based on the catalog strength of the wire rope.
- Designed for use with mobile cranes. Can be used to terminate high performance, rotation resistant ropes, and standard six (6) strand ropes.
- Easy to install assembly utilizes Crosby® WIRELOCK® socketing compound.
- Sockets and buttons are reusable.
- Replacement buttons and sockets are available.
- Locking feature available to prevent rotation of rope.
- Button contains cap with eye that can be attached to, and used to pull rope during reeving process.



SB-427 Button Spelter Sockets

Wire Rope Size		SB-427 Stock No.	Ultimate Load (t)	Weight Each (lbs.)	Socket Only Stock No.	Button Only Stock No.	Dimensions (in.)									
(in.)	(mm)						A	B	C	D	E	F	J	K	L	M
1/2 - 5/8	13-16	1052005	27	6.1	1052107	1052309	7.94	3.23	1.28	1.19	1.22	.57	1.50	3.50	.25	2.93
5/8 - 3/4	16-19	1052014	45	10.3	1052116	1052318	9.44	3.88	1.53	1.38	1.44	.66	1.75	4.28	.38	3.43
3/4 - 7/8	19-22	1052023	57	17.1	1052125	1052327	10.81	4.41	1.78	1.62	1.69	.75	2.06	4.78	.38	3.96
7/8 - 1	22-26	1052032	82	29.2	1052134	1052336	12.88	5.48	2.03	2.00	2.00	.89	2.44	5.62	.62	4.52
1-1/8 - 1-1/4	28-32	1052041	136	46.0	1052143	1052345	14.90	5.68	2.53	2.25	2.50	1.11	2.94	7.08	.75	5.72
1-3/8 - 1-1/2	35-38	1052050	161	78.0	1052152	1052354	18.06	7.17	3.03	2.75	2.75	1.24	3.62	8.08	.75	6.76

SB-427TB (Bolt, Nut and Cotter Pin)

Wire Rope Size		SB-427 Stock No.	Ultimate Load (t)	Weight Each (lbs.)	Socket Only Stock No.	Button Only Stock No.	Dimensions (in.)									
(in.)	(mm)						A	B	C	D	E	F	J	K	L	M
1/2 - 5/8	13-16	1052406	27	6.1	1052107	1052309	7.94	3.23	1.28	1.19	1.22	.57	1.50	3.50	.25	2.93
5/8 - 3/4	16-19	1052415	45	10.3	1052116	1052318	9.44	3.88	1.53	1.38	1.44	.66	1.75	4.28	.38	3.43
3/4 - 7/8	19-22	1052424	57	17.1	1052125	1052327	10.81	4.41	1.78	1.62	1.69	.75	2.06	4.78	.38	3.96
7/8 - 1	22-26	1052433	82	29.2	1052134	1052336	12.88	5.48	2.03	2.00	2.00	.89	2.44	5.62	.62	4.52
1-1/8 - 1-1/4	28-32	1052442	136	46.0	1052143	1052345	14.90	5.68	2.53	2.25	2.50	1.11	2.94	7.08	.75	5.72
1-3/8 - 1-1/2	35-38	1052451	161	78.0	1052152	1052354	18.06	7.17	3.03	2.75	2.75	1.24	3.62	8.08	.75	6.76

Wirelock Requirements

Wire Rope Size		WIRELOCK® Required (cc)	WIRELOCK® Stock No.	WIRELOCK® Kit Size (cc)
(in.)	(mm)			
1/2 - 5/8	13-16	35	1039602	100
5/8 - 3/4	16-19	60	1039602	100
3/4 - 7/8	19-22	100	1039602	100
7/8 - 1	22-26	140	1039602*	100
1-1/8 - 1-1/4	28-32	250	1039604	250
1-3/8 - 1-1/2	35-38	420	1039606	500

* 2 kits required.

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Button Spelter Sockets

CROSBY® SPELTER BUTTON SB-427B APPLICATION INSTRUCTIONS



This procedure is provided to give instructions for installation of wire rope into the Crosby® SB-427B Spelter Button using **WIRELOCK®** socketing material or zinc socketing material. **Additionally, instructions regarding the reuse of spelter buttons are included.** The spelter button is part of a socket assembly that includes a socket basket, pin, cotter pin and button. If there are any questions regarding these instructions, please contact The Crosby Group LLC at (918) 834-4611 and request technical assistance.

NOTE: Many high performance ropes require special attention to prevent rope damage during cutting, seizing and brooming in preparation for the speltering operation. Attention to the special instructions is required to insure proper termination efficiency. Consult rope manufacturer for specific details.

Installation:

Install button on the rope so that the live end of the rope extends out of small diameter of the button. Broomed end of rope should be inserted to the "MAX FILL" line marked on the button to ensure correct length of engagement with socketing material.

Socketing Using WIRELOCK® Resin Material:

Seizing, cleaning, brooming and preparation of wire rope and pouring of **WIRELOCK®** is to be carried out per instructions provided in the ***WIRELOCK®*** *Technical Data Manual,* and ***WIRELOCK®*** *Warnings and Application Instructions* located on the **WIRELOCK®** Product or in the Crosby® General Catalog.

Socketing Using Zinc Spelter Material:

Seizing, cleaning, brooming and preparation of the wire rope, and pouring of zinc is to be carried out in accordance with recommendations of the *Wire Rope User's Manual, 3rd Edition, Appendix C,* or other approved procedures.

Before operation of the wire rope assembly, it is recommended that all poured sockets, whether with zinc or resin, be proof loaded to seat the cone.

Reuse of Crosby® Spelter Buttons:

The following are general guidelines for the reuse of a Crosby® SB-427B Button. The use and inspection of used buttons are the responsibility of the user.

Procedure For Removing Spelter Cone

- Cut the rope close (½") to the nose end of the button and press the cone out of the button.
- For metallurgical, medical and environmental reasons, we do not recommend the use of heat to remove the spelter cone.
  - However, if this is the only means available for removing the zinc cone, care should be taken not to exceed 850°F (450°C) surface temperature. The preferred method would be a slow heat in a temperature controlled oven. If a torch (rosebud) is used, the heat spot shall be monitored with a tempil stick or a temperature indicator to prevent localized heating from exceeding the 850°F (450°C) limit.
  - To remove a **WIRELOCK®** cone, heat the surface of the button to 350˚F (177°C). Do not exceed the 850˚F (450°C) limit for any localized hot spot. Leave for 5-10 minutes, then drive the cone out with a hammer and drift.

Selection of Buttons For Reuse

- Use only buttons that:
  - Do not show discoloration from excessive heating.
  - Do not show any signs of welding.
  - Select only buttons that have been cleaned and have passed a Magnetic Particle Inspection by a qualified technician (Level II ASNT-SNT-TC-1A-88) per ASTM E709. Acceptance criteria shall be per ASTM E125, Types II-VIII, Degree 1. No cracks are acceptable.
  - Select only buttons that do not show any signs of overloading or wear.
  - Select buttons that are free from nicks, gouges and abrasions. Indications may be repaired by lightly grinding until surfaces are smooth, provided they do not reduce the dimensions by more then 10% of the nominal catalog dimension.
  - Select buttons that are not distorted, bent or deformed.

NOTE: Buttons having any of the indications as outlined above shall not be reused.

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DO NOT BE FOOLED...

You could be putting lives and your business at risk for a few pennies.

There are suppliers who would like you to believe swaging sleeves are a commodity. Don't be fooled, IT'S NOT SO!

The Crosby COLD-TUFF® process is the difference.



- Since 1973, Crosby's proprietary COLD-TUFF® heat treat process has made swaging sleeves more reliable during the demanding conditions of swaging.
- COLD-TUFF® virtually eliminates the cracking of sleeve during the swaging process, especially at extremely low temperatures.
- The steel microstructure created from the COLD-TUFF® heat treatment process reveals a uniform structure in the Crosby sleeve, that result in Charpy impact values averaging up to ten times greater than those of competitive sleeves. See photo lower left.
- The improved steel microstructure provides increased resistance to cracks that can occur during the sling building process. Cracks are initiated at stress risers.
  - Crosby COLD-TUFF® sleeves dramatically reduce the effects of stress risers that can occur during swaging, where the "cold worked" sleeve makes contact with the wire. Stress risers are not detectable through visual inspection. See photo at right.
  - Because of the higher Charpy values found in Crosby COLD-TUFF® sleeves, stress risers caused by stenciling are not detrimental.

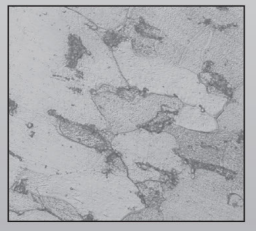


COLD-TUFF® process results in superior ductility that reduces the effects of stress risers.

Un-retouched Photomicrographs of Sleeve (Original Magnification @ 500x)



Crosby COLD-TUFF® Heat Treatment
The ***uniform*** microstructure of the COLD-TUFF® process results in superior ductility and toughness properties.



Competitor's Heat Treatment
The ***non-uniform*** microstructure may lack proper ductility and toughness required for swaging and use.

The Mark Of Quality Is On Every National Sleeve

Crosby or CG and COLD-TUFF are clearly stenciled on every sleeve we make (legible even after swaging) leaving no question as to the quality or the manufacturer.

Crosby COLD-TUFF® and National Swage Fittings
Reliability You Can Depend On

Crosby®

www.thecrosbygroup.com

AWRF Est. 1976

P.O. Box 3128, Tulsa, OK 74101 • E-mail: crosbygroup@thecrosbygroup.com • (918) 834-4611, Fax: (918) 832-0940

WIRELOCK®



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WIRELOCK® Socketing Compound

WIRELOCK® is a unique socketing compound for use with wire ropes. The product eliminates the hazards of working with a molten metal, and allows socketing to be done on site, when required, in virtually all weather conditions.

WIRELOCK® is generally used in long term applications such as crane ropes, boom pendants, winches, lifting straps, mast stays, suspended roofs, bridges, mining and offshore mooring systems.

Efficiency

When fabricated properly, **WIRELOCK®** wire rope assemblies are 100% efficient based on the minimum breaking force of the wire rope. **WIRELOCK®** is designed to be used with steel wire rope, galvanized wire ropes and stainless steel wire ropes.

Installation Information

Socketing Using WIRELOCK® Resin Material:

Seizing, cleaning, brooming and preparation of wire rope and pouring of **WIRELOCK®** is to be carried out per instructions provided in the **WIRELOCK®** Technical Data Manual located in this manual on Page 73-77, and the **WIRELOCK®** Warnings and Application Instructions located on the **WIRELOCK®** Product or in the Crosby® General Catalog.

Before operation of the wire rope assembly, it is recommended that all poured sockets, whether with zinc or resin, be proof loaded to seat the cone.

Facts about WIRELOCK®



1. **WIRELOCK®** is designed to gel (change from a liquid to a solid), in approximately 20 minutes at 65°F (18°C). To ensure that the kits are not adversely affected by storage, they should be stored in a dry place at a temperature of between 50°F and 75°F (10°C and 24°C) and away from any source of



direct heat. **WIRELOCK®**, like all polyester resins, is temperature sensitive. An increase in temperature of 15°F (10°C) shortens the gel time by approximately 50%. A decrease in temperature of 15°F (10°C) lengthens the gel time by approximately 100%.

2. KIT SIZES

   100 cc
   250 cc
   500 cc
   1000 cc
   2000 cc

   Other sizes available to order.

   The specific gravity of **WIRELOCK®** is 1.73. Therefore, 1000 cc's will weigh 1.73 kilos or 3.81 lbs. 250 cc's will weigh

   $$\frac{1.73 \times 250}{1000} = 0.43 \text{ kilos or } 0.95 \text{ lbs.}$$

3. **WIRELOCK®** Wire Rope Assemblies are 100% efficient when used with steel wire rope, galvanized wire ropes and stainless steel wire ropes. We do not advise the use of stainless steel wire rope in a salt water marine environment without regular inspection. In the presence of an electrolyte, i.e., sea water, electrolytic degradation of the stainless steel wire rope can occur. This phenomenon, known as crevice corrosion, will impair the integrity of the rope in the region near to the neck of the socket. Crevice corrosion also occurs when white metal is used for socketing. (Zinc should not be used to socket stainless steel rope.) However, the onset of crevice corrosion in resin sockets appears to be faster than when white metal is used. Other rope types do not exhibit this behavior.

4. **WIRELOCK®** is approximately 20% the weight of zinc.
5. The strength of **WIRELOCK®**, in its cured state, is not

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WIRELOCK® Socketing Compound



Typical example of the swelling of stainless steel rope due to crevice corrosion

adversely affected by cold temperatures.

6. **WIRELOCK®** must be mixed and poured (see 6.3) within the temperature range of 27°F - 95°F (-3°C - 35°C). The kits are not adversely affected by storage at temperatures below 27°F (-3°C). It is recommended the **WIRELOCK®** kit be stored in a cool place.
7. The operating temperature of **WIRELOCK®** is +240°F to -65°F (+115°C to -54°C).
8. When cured, **WIRELOCK®** has a hardness of approximately 55 Barcol. When the resin has set fully (opaque green or mustard color) only a slight scratch mark will be seen when a sharp object, such as a screwdriver blade, is scraped over the surface of the resin. On a small socket, it is quite normal to have a very thin tacky layer on the surface of the resin. The scratch test can be carried out through this layer.
9. Cracks which may appear on the top of the cured cone are surface crazing only, and are the result of heat stresses and shrinkage upon a thin layer of unfilled resin covering the tops of the wires. The crazing does not affect the strength of the termination within the socket.
10. Shrinkage of the **WIRELOCK®** cone may leave a gap between the cone and the socket wall. This is normal, particularly with large sockets and high ambient temperatures. This in no way affects the efficiency of the assembly. Upon loading, the cone will be seated perfectly in the socket. The shrinkage of **WIRELOCK®** is between 1.5 - 2.0%. In high volume **WIRELOCK®**, the shrinkage is about 0.5%.
11. Excessive numbers of horizontal rings in the socket may increase the load required to “seat” and wedge the cone within the socket. They should be avoided whenever possible, and a proof load applied (not exceeding 60% of MBL) if they must be used. Alternatively, they should be filled in with clay prior to placing the socket on the rope.
12. **WIRELOCK®** poured sockets should not be used in environments of strong caustic or acid solutions. **WIRELOCK®** is not affected by oils, or grease or salt water.
13. **WIRELOCK®** is, by design, a compressive resin. Therefore, when removed from the socket a **WIRELOCK®** cone, if hit by a hammer, may shatter. In a socket, even under extreme loads or shockloads, the **WIRELOCK®** cone remains solid and 100% efficient.
14. The shelf life of **WIRELOCK®** is eighteen (18) months (check label before use) from the date of manufacture.

Worldwide Agency Approvals

- ✓ Lloyds Register of Shipping
- ✓ Det Norske Veritas
- ✓ American Bureau of Shipping
- ✓ United States Coast Guard
- ✓ Registro Italiano Navale
- ✓ Germanischer Lloyd

STANDARD CONSIDERATIONS

ASME B30.9

All slings terminated with sockets shall be proof loaded. The proof load shall be a minimum of 2 and a maximum of 2-1/2 times the single leg vertical hitch rated load. The proof load should be that specified by the wire rope or fitting manufacturer’s recommendation provided that it is within the above specified proof load range. When sockets are used in sling assemblies the minimum recommended design factor is 5.

ASME B30.5

Cranes require that boom pendants terminated with sockets be proof tested. When sockets are used in boom pendant lines the minimum recommended design factor is 3.

See OSHA and ASME for full information.

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WIRELOCK®

RESIN FOR SPELTER SOCKETS
NOT AVAILABLE IN CANADA
Note: For use on 416, 417, 427 and 517 spelter sockets only.

- 100% termination efficiency.
- Temperature operating range is -65° F to +240° F (-54°C to +116°C).
- Ideal for on site applications.
- No hazardous molten metal.
- Improved fatigue life.
- Pouring temperature without booster pack is 48° F to 110° F (6.67°C to 43.3°C).
- One booster pack if pouring temperature is 35° F to 48° F (1.67°C to 8.89°C).
- Two booster packs if pouring temperature is 27° F to 35° F (-2.78°C to +1.67°C).
- Refer to WIRELOCK® Technical Manual for more information.





SEE APPLICATION AND WARNING INFORMATION

Para Español: www.thecrosbygroup.com *On Pages 94-95*

Approvals:

Lloyds Register of Shipping
Det Norske Veritas (DNV)
United States Coast Guard
United States Navy
American Bureau of Shipping
ISO 17.558
DNV-OS-E304

NATO Numbers:

100 cc	8030-21-902-1823
250 cc	8030-21-902-1824
500 cc	8030-21-902-1825
1000 cc	8030-21-902-1826

Witnessed and tested by American Bureau of Shipping. (ABS)

Approximate U.S. Measurements:
250 cc's Kit 1 Cup

WIRELOCK® W416-7 Socket Compound

W416-7 Kits				Booster Pak Stock No.
Kit Size	Kit Per Case	Stock No.	Weight Each (lbs.)	
100	20	1039602	.62	1039603
250	12	1039604	1.25	1039605
500	12	1039606	2.54	1039607
1000	12	1039608	4.59	1039609
2000	12	1039610	9.00	1039611

Guide to amount of WIRELOCK® Required

Wire Rope Size		WIRELOCK® Required (cc)	Wire Rope Size		WIRELOCK® (cc)
(in.)	(mm)		(in.)	(mm)	
1/4	6-7	9	1-3/4	44	700
5/16	8	17	1-7/8	48	700
3/8	9-10	17	2	51	1265
7/16	11	35	2-1/8	54	1265
1/2	13	35	2-1/4	56	1410
9/16	14	52	2-3/8	60	1410
5/8	16	52	2-1/2	64	1830
3/4	20	86	2-5/8	67	1830
7/8	22	125	2-3/4	70	2250
1	26	160	3	76	3160
1-1/8	28	210	3-1/4	82	3795
1-1/4	32	350	3-1/2	88	4920
1-3/8	36	350	3-3/4	94	5980
1-1/2	40	420	4	102	7730
1-5/8	42	495	—	—	—

Wirelock is a hazardous material regulated by US DOT, ICAO/IATA and IMO for transportation.

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DETAILED INSTRUCTIONS FOR THE USE OF WIRELOCK®

With Strand or General Purpose Wire Rope

These instructions explain the proper use of WIRELOCK® for socketing wire rope terminations. When reading and following these instructions, pay close attention to warnings and safety information presented in bold print.

For maximum salety and efficiency, use **WIRELOCK®** only as instructed.

1. Warning on Correct Application of WIRELOCK®

It is very important when deciding upon the use of **WIRELOCK®** to note the following:

⚠ WARNING

- **Incorrect use of WIRELOCK® can result in an unsafe termination which may lead to serious injury, death, or property damage.**
- **Do not use WIRELOCK® with stainless steel rope in salt water environment.**
- **Use only soft annealed iron wire for seizing.**
- **Do not use any other wire (copper, brass, stainless, etc.) for seizing.**
- **Never use an assembly until the WIRELOCK® has gelled and cured.**
- **Remove any non-metallic coating from the broom area.**
- **Non Crosby sockets with large grooves need to have those grooves filled before use with WIRELOCK®.**
- **Read, understand and follow these instructions and those on product containers before using WIRELOCK®.**

2. Safety and Health Precautions for Using WIRELOCK®

It is very important that certain precautions be taken when using **WIRELOCK®** for a wire rope socket termination. When using the product, be sure to read information on product containers and note the following:

⚠ CAUTION

- **WIRELOCK® resin, in liquid state, is flammable.**
- **Chemicals used in this product can give off toxic fumes and can burn eyes and skin.**
- **Never use out-of-date material.**
- **Use only in well ventilated work areas.**
- **Never breathe fumes directly or for extended time.**
- **Always wear safety glasses to protect eyes.**
- **Always wear gloves to protect hands.**
- **Avoid direct contact with skin anywhere.**

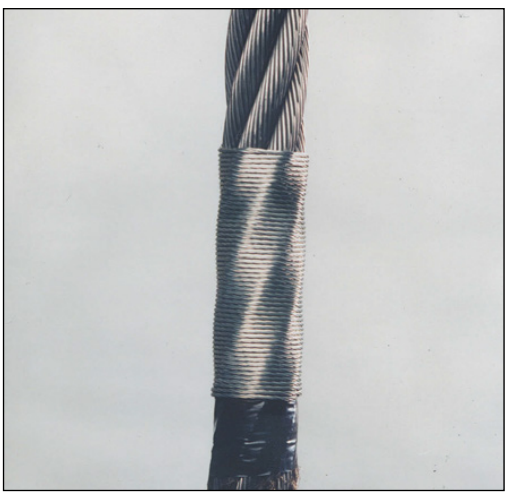
3. Selection of Socket

3.1 WIRELOCK® is recommended for use with sockets that comply with Federal or International (CEN, ISO) Standards.

3.2 WIRELOCK® as with all socketing media, depends upon the wedging action of the cone within the socket basket to develop full efficiency. A rough finish inside the socket may increase the load at which seating will occur. Seating is required to develop the wedging action.

3.3 Measure the rope ends to be socketed. The rope end should be of sufficient length so that the ends of the unlaid wires (from the strands) will be at the top of the socket basket.

3.4 Next, apply the seizing one (1) socket basket length from the end of the rope minus one (1) rope diameter. The length of the seizing must be at least two (2) rope diameters long. Additional information can be secured from your Wire Rope Users Manual or your Wire Rope Manufacturers' Catalogs or National Standard. Seizing wire should be a soft annealed iron wire.



Seizing of Wire Rope

3.5 Plastic coated or plastic filled wire ropes must have all plastic material (non-metallic materials) removed from within the broomed area.

3.6 The socket basket should be examined prior to use and loose scale, dirt or grease removed.

Do not use oversized sockets for Wire Rope.

3.7 When socketing Strand, the time honoured method of one size up when choosing the socket is generally still applicable in the vast majority of cases. However, caution should be exercised as tests have shown that the length of the socket basket should be five (5) times the strand diameter or fifty (50) times the maximum wire diameter, whichever is the greater.

3.8 **Inserting the broom into the socket.**

There are two procedures that can be used to position the broom within the socket. The rope

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